Practice MCQ
MTH166

Q1. The general solution of the equation $y'' + 16y = x^2$ is

(a) A cos x + B sin x + $4x^2$

(b) A cos 4x + B sin 4x + $4x^2 - \frac{1}{2}$

(c) A cos 4x + B sin 4x + 4x

(d) None of these

Q2. The Particular integral of the differential equation $(D^2+4D+4)y = 2\cosh x$ is

(a) $\frac{1}{9}e^{-x}+e^{x}$ **(b)** $\frac{1}{9}e^{x}+e^{-x}$ **(c)** $e^{x}+\frac{1}{9}e^{-x}$ **(d)** None of these

Q3. Solution of $x^2y'' - 3xy' + 4y = 0$ is

(a) $(A+Bx)e^{2x}$ (b) $(c_1 + c_2 \log x)x^2$ (c) $(c_1 + c_2 \log z)z^2$ (d) $(c_1 + c_2 z)logz$

Q4. If f (D) y = xe^{2x} then its particular integral is given by

(a) $e^{2x} \frac{1}{f(D+2)} x$ **(b)** $x \frac{1}{f(D+2)} e^{2x}$ **(c)** $x \frac{1}{f(D-2)} e^{2x}$ **(d)** none of these

Q5. The particular integral of the differential equation $y'' + 2y' + 3y = 1 + e^x$ is

(a) $\frac{1}{3}(1 + e^x)$ (b) $\frac{1}{3}(1 - e^x)$ (c) $\frac{1}{3} - \frac{1}{6}e^x$ (d) $\frac{1}{3} + \frac{1}{6}e^x$

Q6. Find the value of x of simultaneous differential equation

$$\frac{dx}{dt}+\frac{dy}{dt}+2x+y=0\,,\quad \frac{dy}{dt}+5x+3y=0$$

(a) $x = A\cos t + B\sin t$ **(b)** $x = A\cos y + B\sin y$ **(c)** $x = (A+Bt)\cos t$ **(d)** $x = (A+Bt)\sin t$

Q7. If $(D^2 + 2D+1)y = e^{2x}$ then by the method of undermined coefficient the assumed particular integral is given by

(a) $Kx\, e^{2x}$ (b) Ke^{2x} (c) $Kx^2\, e^{2x}$ (d) $Kx^3\, e^{2x}$

Q8. Which of the following is correct?

(a) $\frac{1}{f(D)} \cos ax = \frac{1}{f(D^2 = -a^2)} \cos ax$, $f(D^2 = -a^2) \neq 0$

(b) $\frac{1}{f(D)} \cos ax = \frac{1}{f(D=a)} \cos ax$, $f(a) \neq 0$.

(c) $\frac{1}{f(D)} \sin ax = \sin ax \frac{1}{f(D+a)}$

(d) none of these

Q9. Solving equations $\frac{dx}{dt} + wy = 0, \frac{dy}{dt} - wx = 0 \ for\, x$, x is equal to

(a) $Ae^{wt} + Be^{-wt}$ (b) $(A + Bt)e^{wt}$ (c) $A\cos wt + B\sin wt$ (d) None of these

Q10. we can write equation $Dx + 2x - 3y = t$ as

(a) $(D+2)x - 3y = t$ (b) $(D-3)y + 2x = t$ (c) $(D+2)x + y = 3t$ (d) None of these

Q11. If differential equation $y' = \frac{3y}{x}$ is normal over the interval I then I is

(a) $(-\infty, 2)$ **(b)** $(-\infty, \infty)$ **(c)** $(-2, \infty)$ **(d)** $(-\infty, 0)$

Q12. What is lowest order of homogeneous linear homogeneous differential equation with constant coefficient whose particular solution is $3\cos 2x + 5\sinh 3x$

(a) 2 (b) 3 (c) 4 (d) 6

Q13. What are the characteristic roots of a homogeneous linear DE with constant coefficients having $4 + xe^{2x}$ as its particular solution?

(a) 0, 2 **(b)** 4, 2 **(c)** 4, 2, 2 **(d)** 0, 2, 2

Q14. . The functions f_1 , $f_2, \_\_\_\_\_\_, f_n$ are said to be linearly dependent if W (f_1, $f_2, \_\_\_\_ f_n$) is

(a) 0 **(b)** $\neq 0$ **(c)** 1 **(d)** None of these

Q15. Particular solution of the problem

$(D^2 + 4)y = 0$; $y(0) = 0$, $y'(\pi) = 2$ is

(a) $y(x) = \sin 2x$ **(b)** $y(x) = \cos 2x$ **(c)** $y(x) = \tan 2x$ **(d)** none of these

Q16. If the roots of the auxiliary equation of a linear differential equation are 4,4 then. its complementary function is

(a) $C_1e^{4x} + C_2e^{4x}$ **(b)** $(C_1 + C_2x)e^{4x}$ **(c)** $(C_1 + C_2x^2)\, e^{4x}$ **(d)** $(C_1x + C_2x^2)e^{4x}$

Q17. Value of $\frac{1}{D^2+a^2}\sin ax =$

(a) $-\frac{x}{2a}\sin ax$ (b) $\frac{x}{2a}\sin ax$ (c) $-\frac{x}{2a}\cos ax$ (d) $\frac{x}{2a}\cos ax$

Q18. The Particular Integral of $\frac{d^2y}{dx^2} + y = \sin x$

(a) $-\frac{x}{2}\sin x$ (b) $\frac{x}{2}\sin x$ (c) $-\frac{x}{2}\cos x$ (d) $\frac{x}{2}\cos x$

Q19. If X is a function of x, then $\frac{1}{D-a}X =$

(a) $e^{-ax}\int e^{ax}X\,dx$ (b) $e^{ax}\int Xe^{-ax}\,dx$ (c) $e^{-ax}\int e^{ax}\,dx$ (d) $e^{ax}\int e^{-ax}\,dx$

Q20. The complementary function of $\frac{d^2y}{dx^2} + y = \sin x$ is

(a) $c_1 \cos x + c_2 \sin x$ (b) $c_1 \cos 2x + c_2 \sin 2x$ (c) $c_1 \cos 2x + c_2 \sin x$ (d) $\sin x$

Q21. The general solution of differential equation $\sin px \cos y - \cos px \sin y = p, \quad where\ p = \frac{dy}{dx}$ is given by

(a) $y = cx + \frac{a}{c^2}$ (b) $y = cx - e^c$ (c) $(y - cx)^2 = a^2c^2 + b^2$ (d) $y = cx - \sin^{-1} c$